

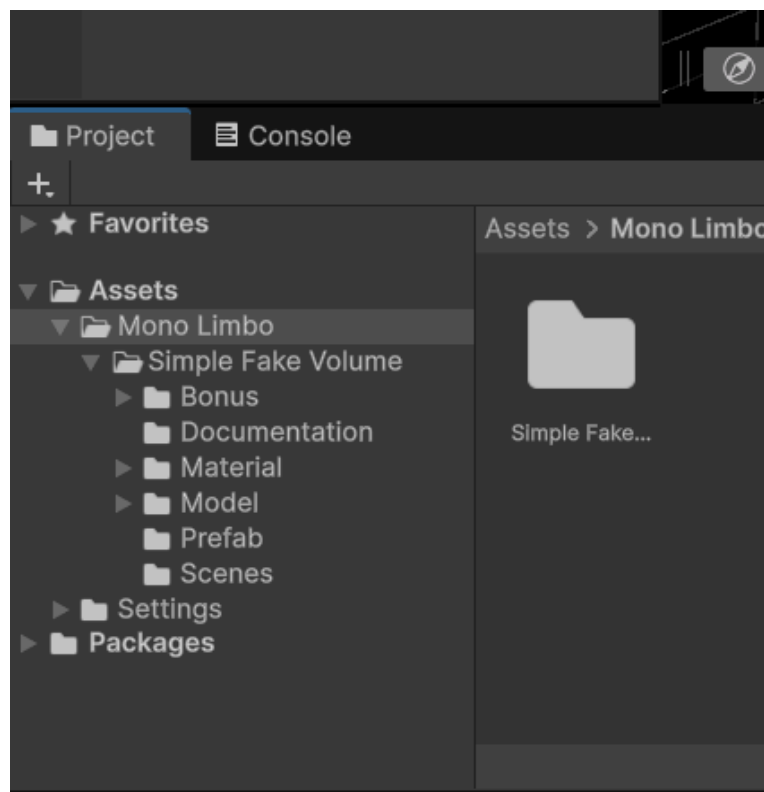
Simple Fake Volume Fog - Documentation

Overview Achieve beautiful, lightning-fast volumetric lighting in your project. Built entirely with standard Shader Graph for maximum optimization, this asset provides a drag-and-drop solution for faking volumetric light beams and point lights without the heavy rendering cost of true volumetrics.

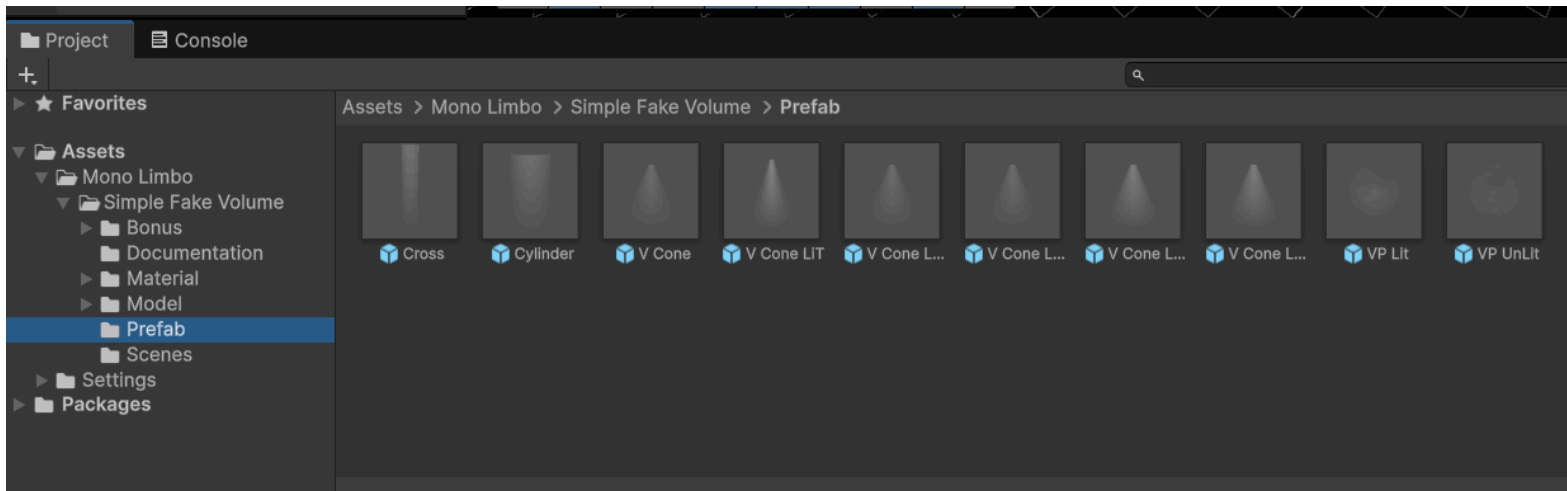
Note: This asset relies on a highly optimized, standard UV/Shader Graph solution.

1. Package Contents

- **Prefab:** 10 ready-to-use prefabs for instant setup (e.g., [V Cone](#), [V Cone Lit Flicker](#), [VP Lit](#), [VP Unlit](#)).
- **Scenes:** 9 ready-to-play sample scenes demonstrating different lighting scenarios (Flash Light, Horror, Street, Party, Passage, etc.).
- **Material:** Contains all core volumetric materials categorized by type (Cone, Point, Colored) along with the raw Shaders and Textures.
- **Bonus:**
 - **FX:** Lens Flares (Default, Normal, Red) and custom SFX (ambient wind loops, lightbulb hum).
 - **Scripts:** Procedural control scripts for dynamic lights and cameras.



2. How to Use (Quick Start)



Basic Setup:

1. Open the **Prefab** folder.
2. Drag and drop the desired volume into your scene hierarchy.
 - Use **VP Lit** or **VP Unlit** for Point Lights.
 - Use **V Cone...** variations for Spotlights.
3. Position the prefab so its origin aligns exactly with your Unity Light source.
4. Scale the transform (X, Y, Z) to match the range and angle of your actual light.

Tweaking the Visuals: Select your volumetric prefab and adjust the **Material** component in the Inspector:

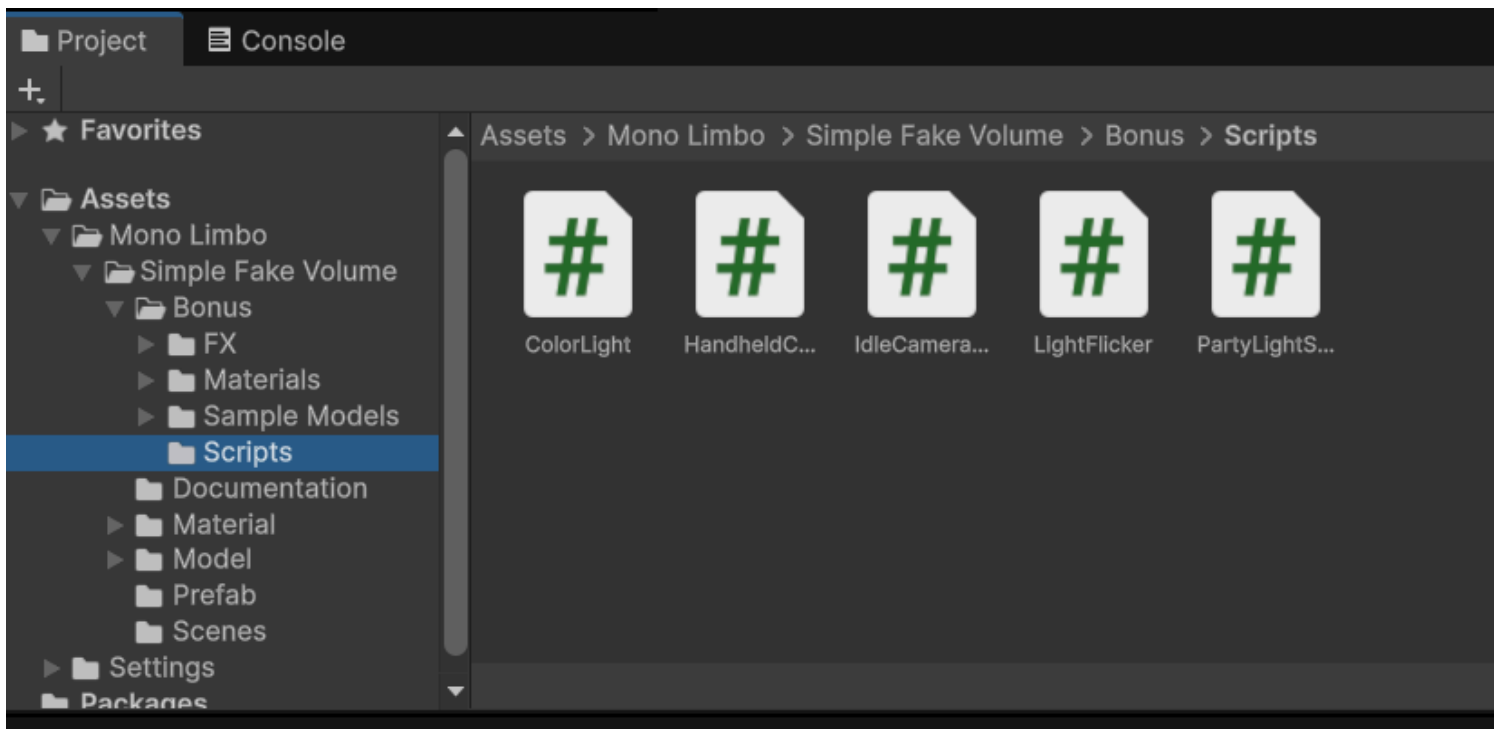
- **Opacity & Color:** Set the density and tint of the beam.
- **Edge & Fade Distance:** Soften intersections with other geometry to prevent harsh clipping into walls, floors, or the camera.
- **Use Wind? / Noise Settings:** Enable noise and adjust Scale, Speed, and Density to simulate scrolling dust particles, atmospheric wind, or underwater light scattering.

3. Procedural Script Toolkit

Included in the **Bonus > Scripts** folder are 5 plug-and-play scripts (**MonoLimboStudio** namespace) that use procedural mathematics (sine waves and Perlin noise) to add organic movement without complex animations.

- **LightFlicker:** Syncs a Unity Light's intensity with a renderer's material opacity to create erratic flickering.
- **PartyLightController:** Continuously cycles RGB colors while flickering. Automatically desynchronizes when duplicated to easily build chaotic nightclub environments.
- **PartyLightSwing:** Simulates the continuous sweeping figure-8 motion of a stage light. Built-in time offsets ensure multiple swinging lights operate independently.
- **HandheldCameraShake:** Adds a realistic handheld feel to the Main Camera by combining a slow "breathing" sway with a fast, nervous jitter.
- **IdleCameraDolly:** Generates a continuous "dolly zoom" (Vertigo effect) by pushing the camera along its Z-axis while inversely adjusting the FOV.

Setup for Scripts: Attach the script to your Light/Camera, assign the target components in the Inspector, and adjust the exposed variables (Speed, Intensity, Color Offsets) to your liking.



4. Shader Graph Breakdown

The core volumetric effect requires zero custom nodes and is built entirely in Unity's standard Shader Graph. It is divided into internal modules for easy modification:

- **Wind Effect & Extra Noise:** Utilizes standard Noise nodes panning over time to simulate environmental atmospheric effects.
- **Edges Fading:** Uses Fresnel logic to soften the hard geometry of the volume mesh, ensuring it renders like a gaseous volume.
- **Camera & Intersection Fading:** Uses Scene Depth and Camera Distance calculations to smoothly fade the fog when clipping into the environment or the player's view.

